

ReactJS HOL- 2

Objective

Create a React application named StudentApp with Home, About and Contact components and render all three components in the main application.

Theory

1. Explain React Components

React components are reusable building blocks of a React application. Each component represents a part of the user interface and can contain its own logic and presentation.

2. Difference Between React Components and JavaScript Functions

React components return JSX to display UI and can receive props or manage state. JavaScript functions perform general programming tasks and are not specifically used to render user interfaces.

3. Types of Components

React provides two main types of components: Functional Components and Class Components. Functional components are simple JavaScript functions, whereas Class Components extend `React.Component` and support lifecycle methods.

4. Explain Class Component

A class component is an ES6 class that extends `React.Component`. It contains a `render()` method and can manage state using `this.state`.

5. Explain Functional Component

A functional component is a JavaScript function that returns JSX. It is lightweight, easy to read, and supports Hooks such as `useState` and `useEffect`.

6. Define Component Constructor

The constructor is a special method used in class components to initialize state and bind methods before the component is rendered.

7. Define `render()` Function

The `render()` function returns the JSX that React displays in the browser. Whenever props or state change, React invokes `render()` again to update the UI.

Prerequisites

- Node.js
- npm
- Visual Studio Code

Implementation Steps

1. Create a React application named StudentApp using `npx create-react-app StudentApp`.
2. Inside the `src` folder, create a `Components` folder.
3. Create `Home.js`, `About.js` and `Contact.js` class components.
4. Display the required welcome message in each component.
5. Import all three components into `App.js`.
6. Render the `Home`, `About` and `Contact` components.
7. Run the application using `npm start`.
8. Open `http://localhost:3000` to verify the output.

Source Code

Home.js

```
import React, {Component} from 'react';
export default class Home extends Component{
  render() {
    return <h3>Welcome to the Home Page of Student Management
Portal</h3>;
  }
}
```

About.js

```
import React, {Component} from 'react';
export default class About extends Component{
  render() {
    return <h3>Welcome to the About Page of the Student Management
Portal</h3>;
  }
}
```

Contact.js

```
import React, {Component} from 'react';
export default class Contact extends Component{
  render() {
    return <h3>Welcome to the Contact Page of the Student
Management Portal</h3>;
  }
}
```

```

    }
  }
}

```

App.js

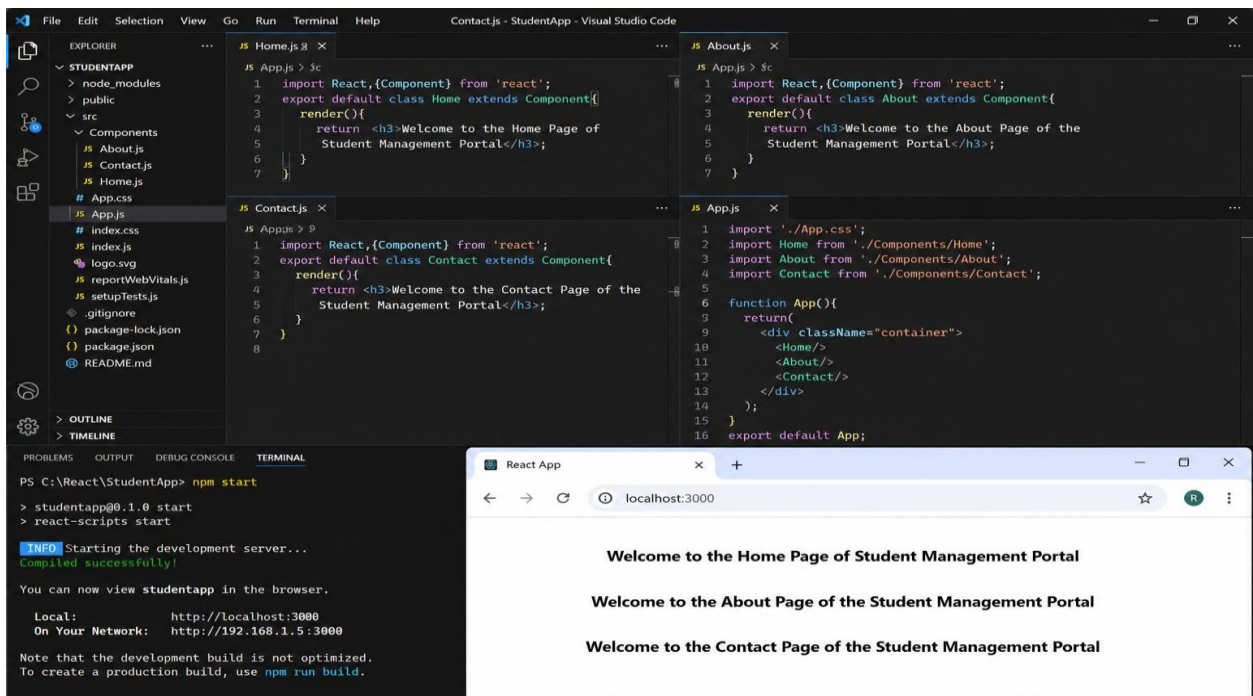
```

import './App.css';
import Home from './Components/Home';
import About from './Components/About';
import Contact from './Components/Contact';

function App() {
  return (
    <div className="container">
      <Home/>
      <About/>
      <Contact/>
    </div>
  );
}
export default App;

```

Output Screenshot:



Learning Outcome

Successfully created multiple React class components, imported them into App.js, and rendered them in a React application.